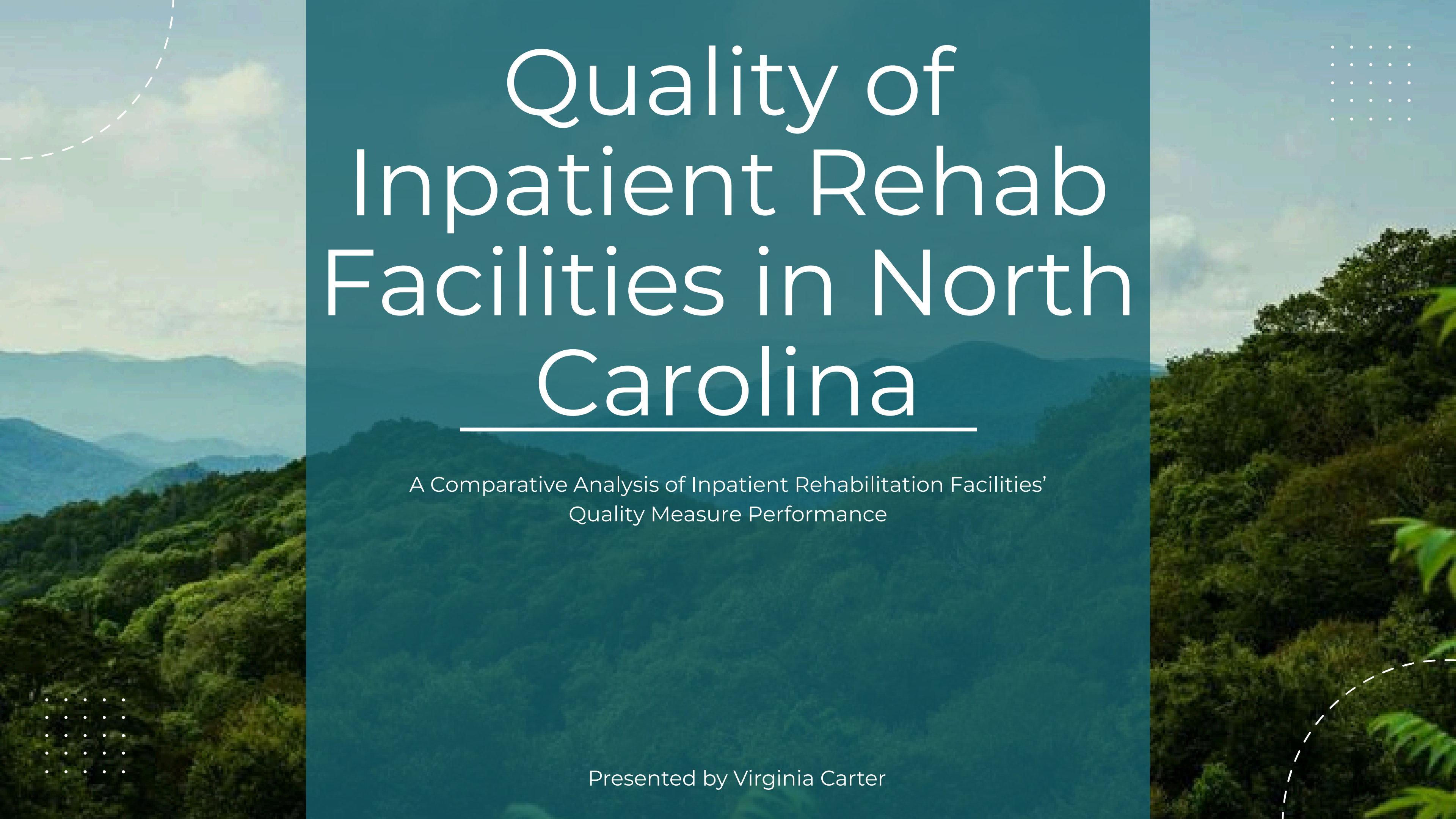


The background of the slide is a photograph of a mountain landscape. In the foreground, there are lush green trees and foliage. In the background, there are rolling mountains under a blue sky with some clouds. A semi-transparent teal rectangle is overlaid on the center of the image, containing the title and subtitle text. There are also decorative white dotted patterns in the corners of the teal area.

Quality of Inpatient Rehab Facilities in North Carolina

A Comparative Analysis of Inpatient Rehabilitation Facilities'
Quality Measure Performance

Presented by Virginia Carter

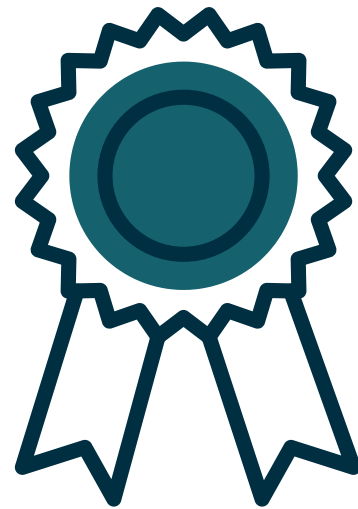
Inpatient Rehabilitation Facility Quality Reporting Program (IRF QRP)

- The IRF QRP was established in 2010 as part of the Affordable Care Act. This program requires Inpatient Rehabilitation Facilities (IRFs) to report quality data to the Centers for Medicare & Medicaid Services (CMS).
- The purpose of this reporting is to ensure quality health care for Medicare beneficiaries through accountability and public disclosure.
- Data is collected from the Patient Assessment Instrument (PAI), the National Healthcare Safety Network (NHSN), and Medicare fee-for-service claims.



KEY FINDINGS

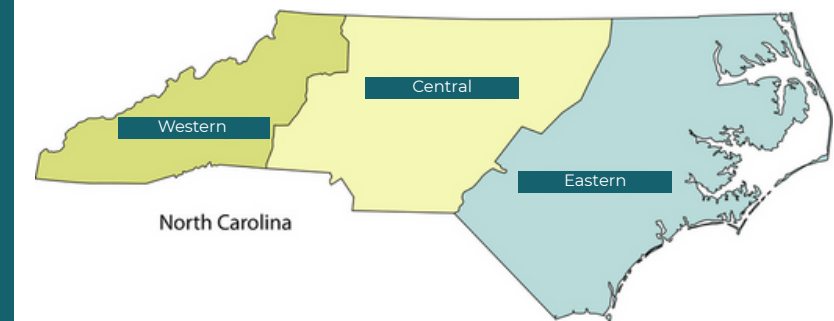
QUALITY



OWNERSHIP
TYPE



REGION



QUALITY

The Quality Reporting Program includes 16 separate measures used to assess performance across inpatient rehabilitation facilities.

This presentation will look at four of these measures that examine effectiveness of care, quality of care, and fall incidents.

01

Discharge to the Community:

- CMS ID: 1019.02
- Discharge to Community–PAC

02

Readmission Rates

- CMS ID: I018.01
- Potentially Preventable Within Stay Readmission Measure

03

Falls:

- CMS ID: I022.01
- Percent of Residents Experiencing One or More Falls with Major Injury

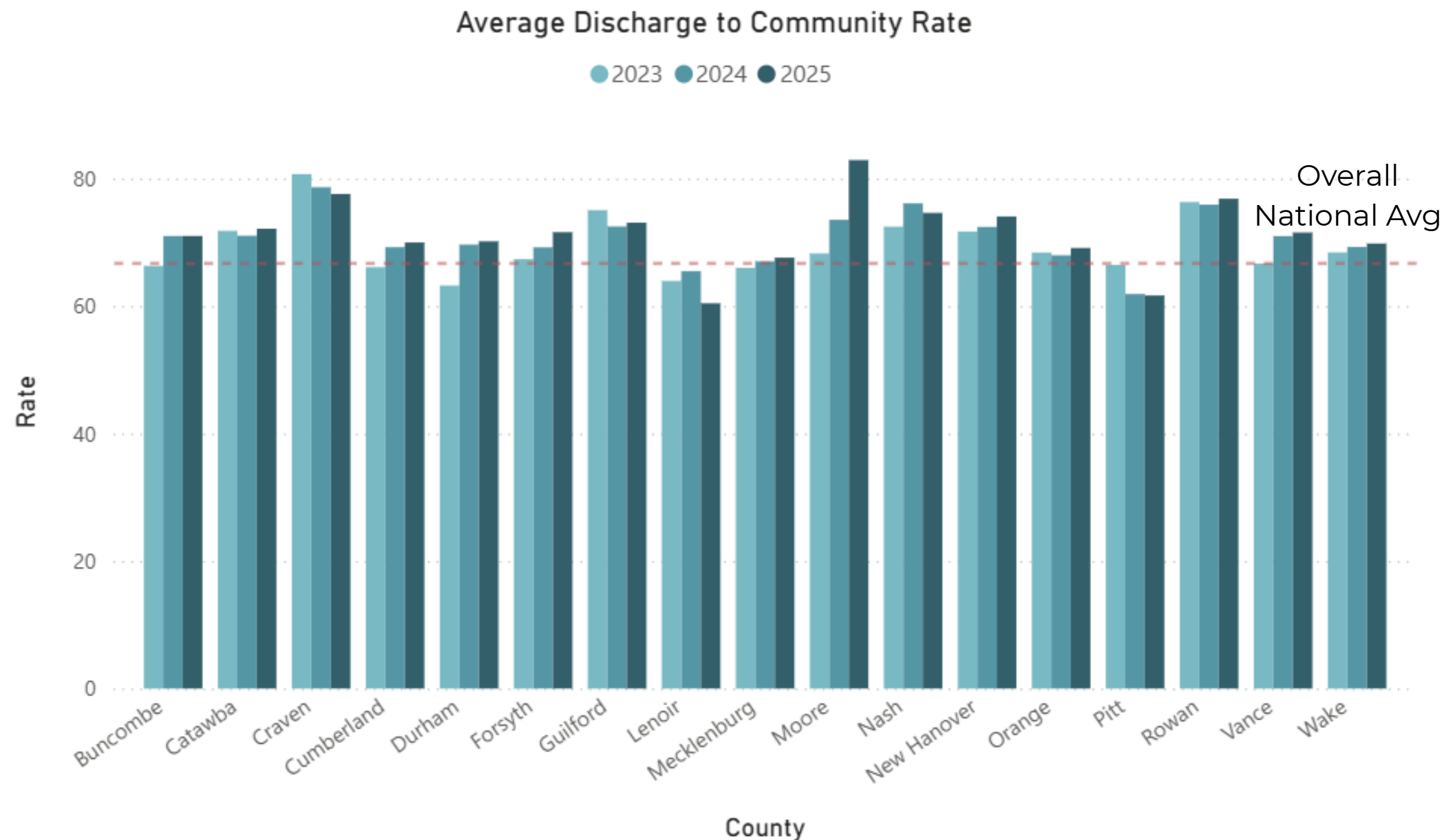
04

Discharge Function Score

- CMS ID: I0.26.01
- Percent of patients who are at or above an expected ability to care for themselves and move around at d/c

DISCHARGE TO THE COMMUNITY

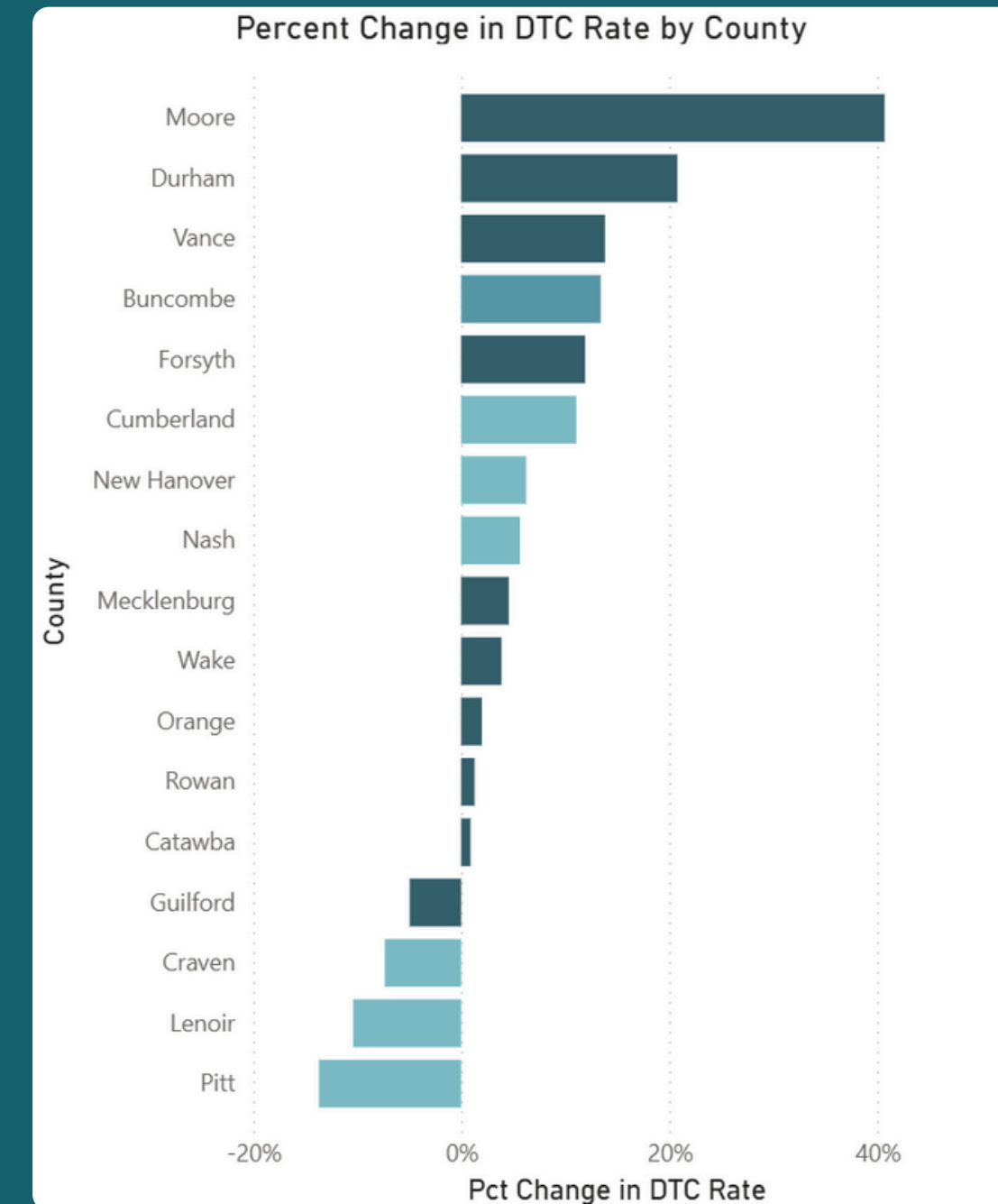
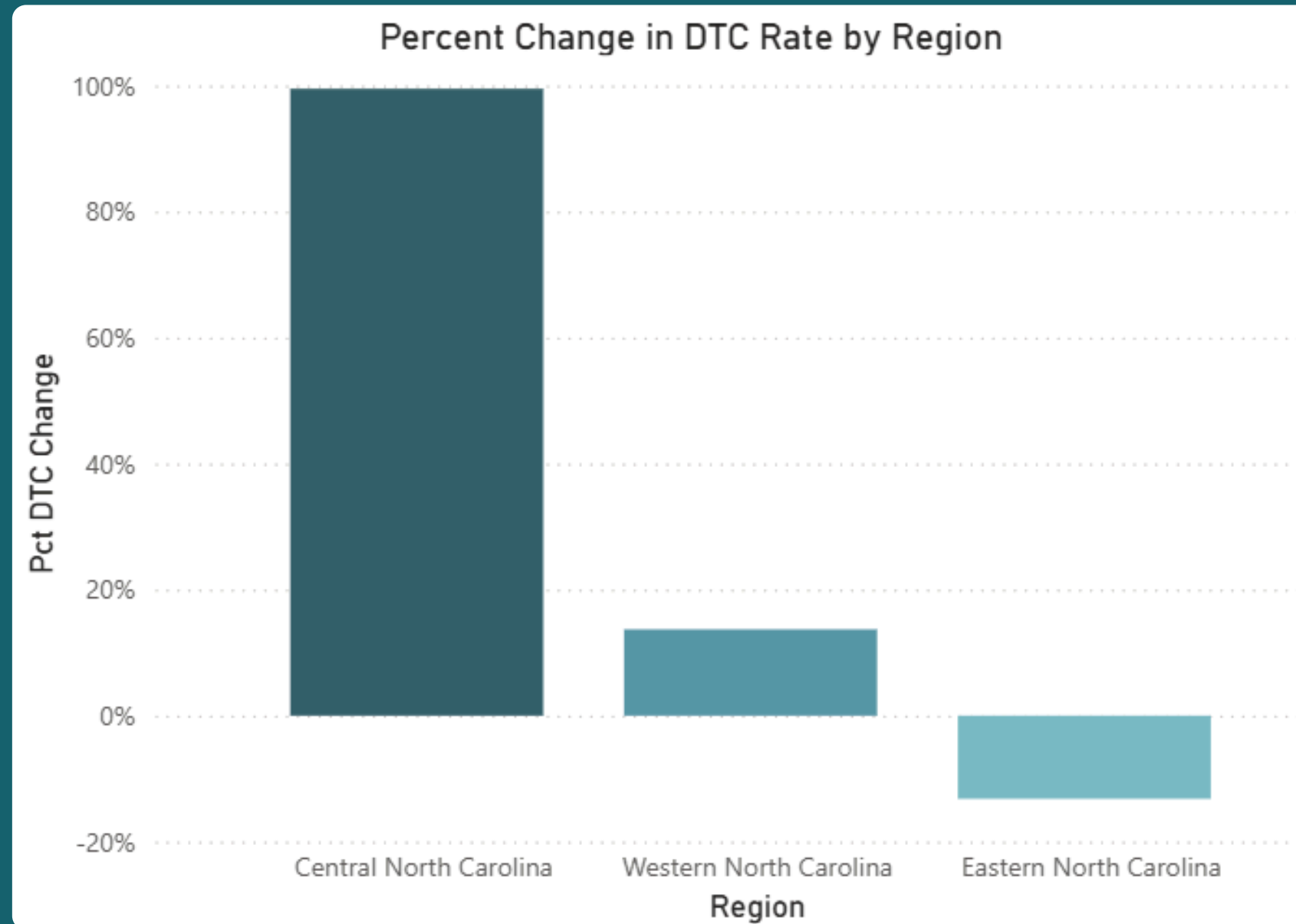
(CMS ID: 1019.02)



- This is a measure of the rate of successful return to home or community from an IRF
- The goal of IRFs is to help patients regain independence after serious illness or injury.
- Success is measured by how many patients safely return to the community instead of another care facility — a higher rate indicates better outcomes.
- For most counties, rates of DTC increased over the three years.
- North Carolina IRFs consistently show higher rates of DTC compared to the national average.

DISCHARGE TO THE COMMUNITY

By Region and County

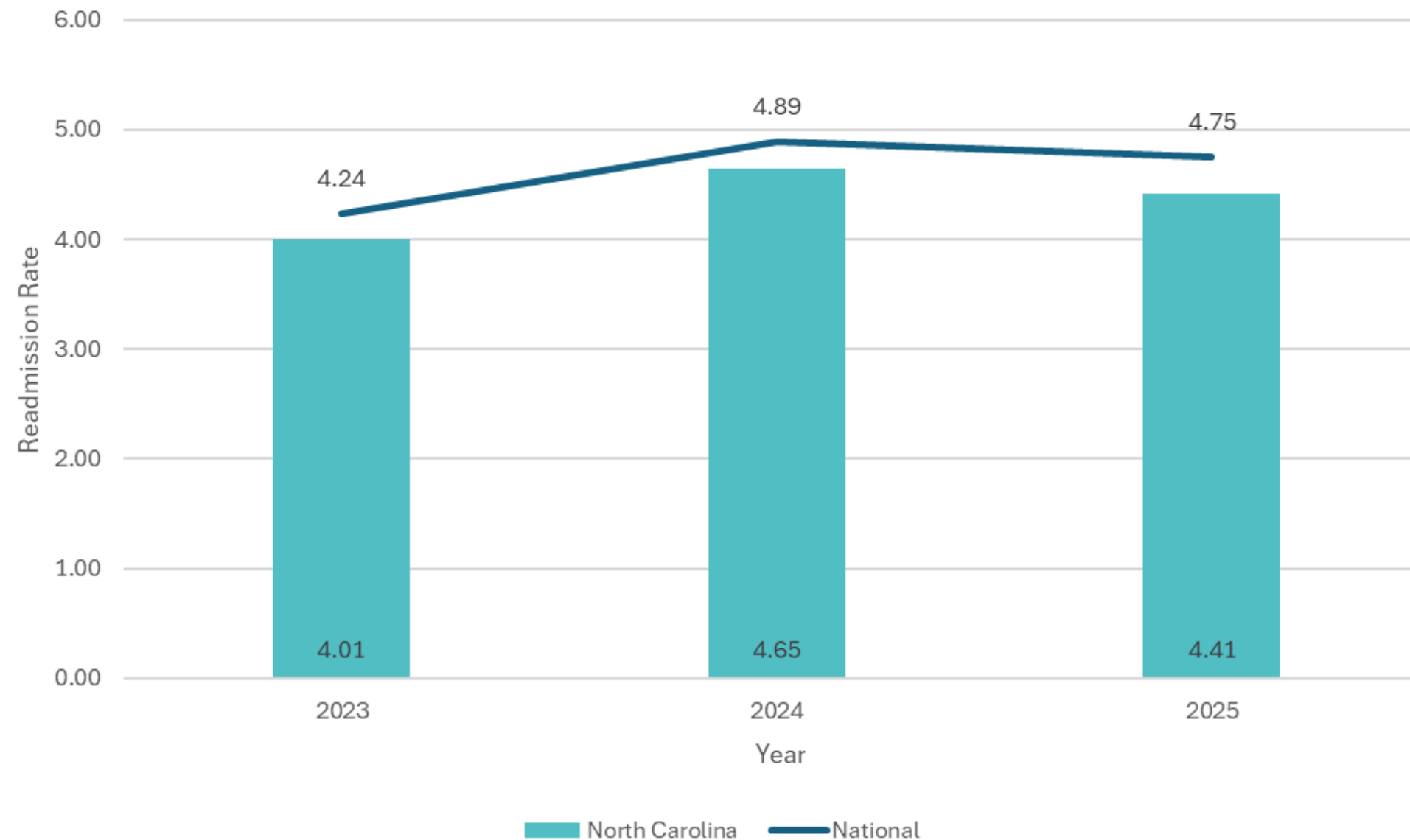


- Eastern North Carolina shows a decrease in their DTC rates, indicating fewer patients are discharging back to the community.
- As seen on the right, the counties showing a decline in rates include Guilford (Central) and Craven/Lenoir/Pitt counties (Eastern)

POTENTIALLY PREVENTABLE *WITHIN STAY* READMISSION MEASURE (CMS ID: I018.01)

- This is a measure of the rate of potentially preventable hospital readmissions *during* the IRF stay.
- 2024 demonstrates a sharp increase in readmission rates compared to 2023, both nationally and in North Carolina.
- North Carolina consistently demonstrate readmission rates lower than the national average, indicating overall good quality of care.
- There was a 16% increase in the rate of readmissions from '23-'24 in NC, compared to a 15% increase nationally.

National Vs. NC Overall Avg Readmission Rates

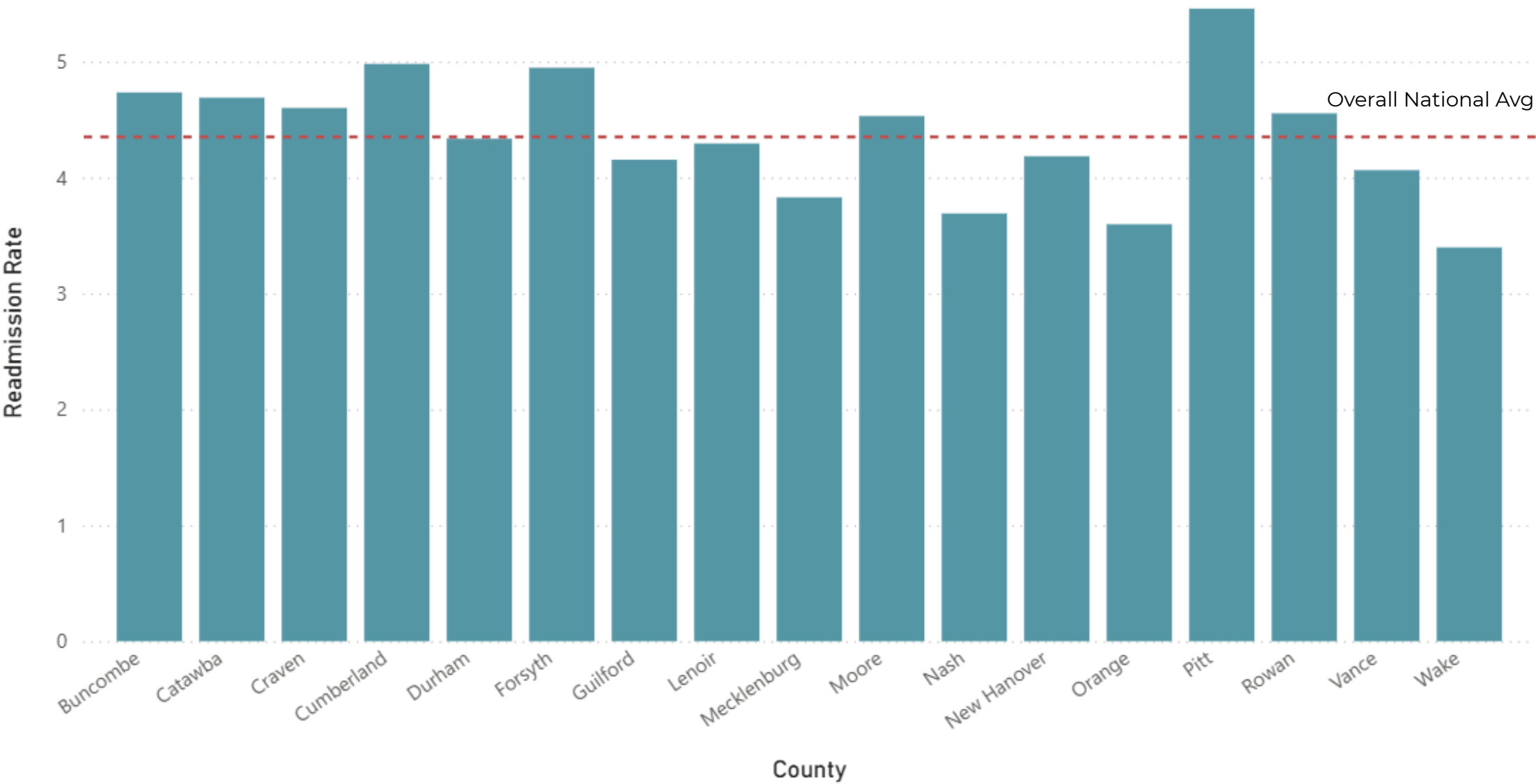


POTENTIALLY PREVENTABLE
WITHIN STAY READMISSION
MEASURE (CMS ID: I018.01)

- A positive % indicates increasing readmission rates
- 47% of NC counties demonstrated an overall average readmission rate greater than the national average.

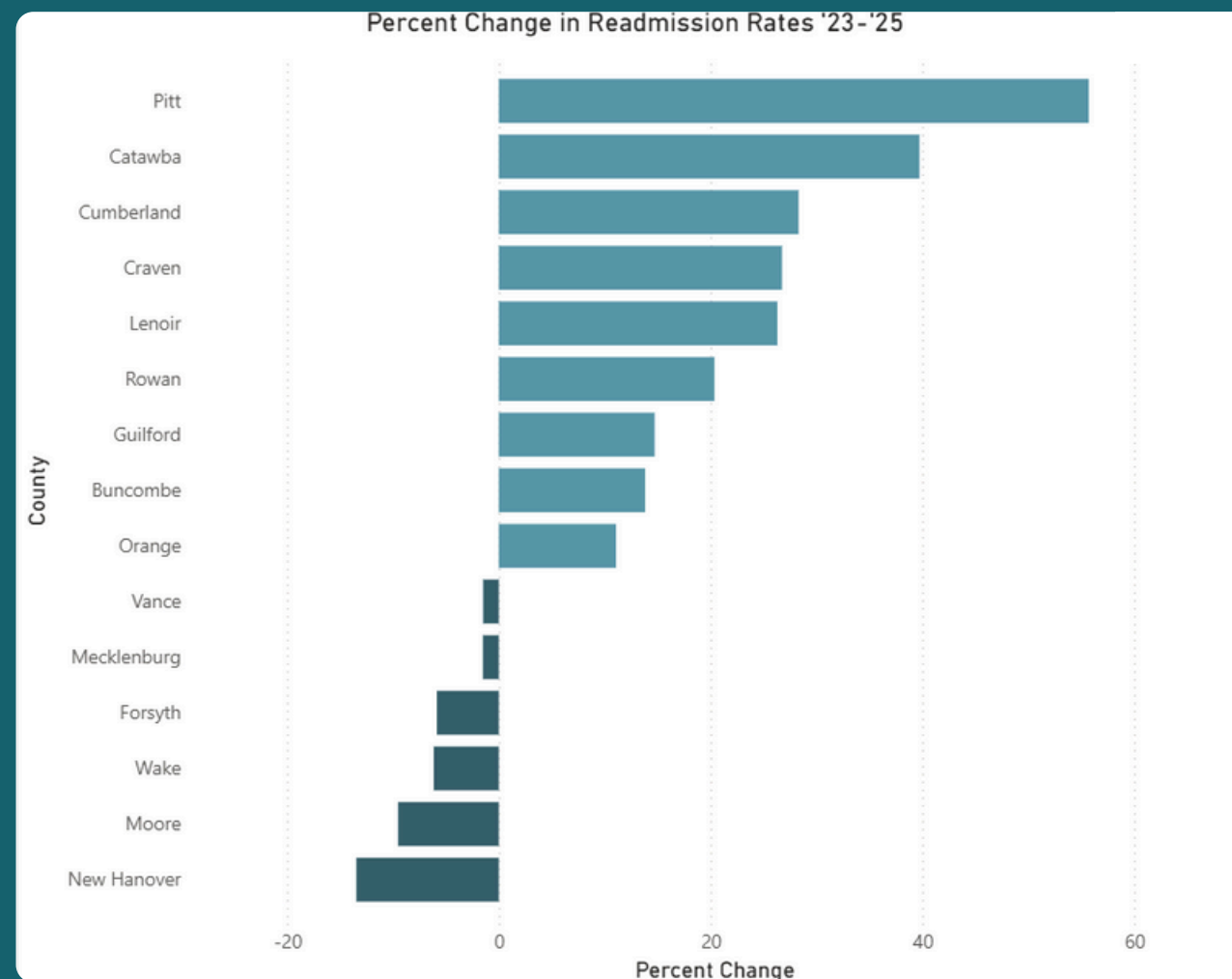
Top 3 Highest Rates	
County	Overall Avg. Readmission Rate
Pitt	5.45
Cumberland	4.98
Forsyth	4.94

NC Overall Avg Readmission Rates

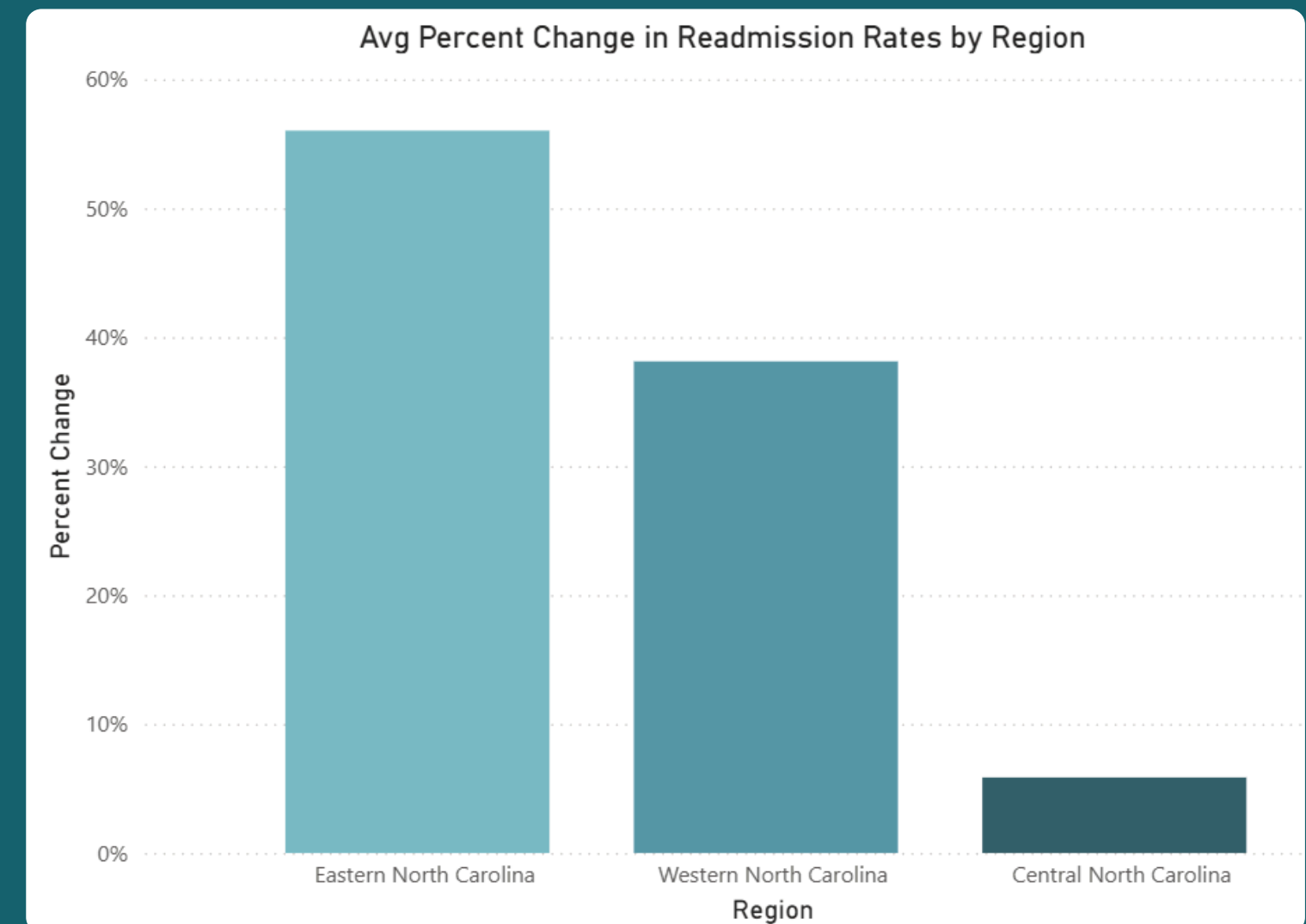


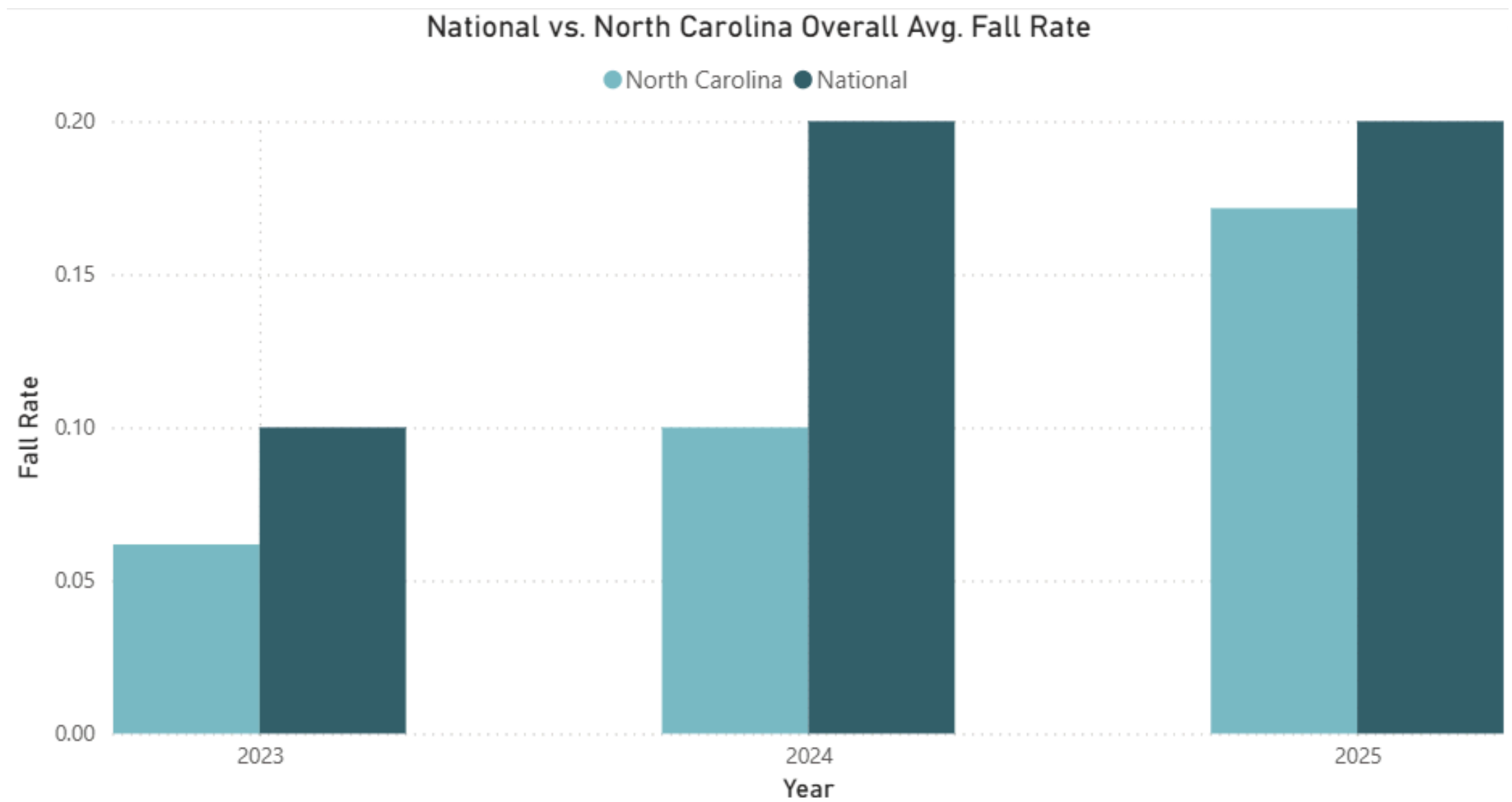
POTENTIALLY PREVENTABLE *WITHIN STAY* READMISSION MEASURE (CMS ID: I018.01)

- Again we see Pitt and Cumberland counties at the top with some of the highest percent in change regarding readmission rates.
- The following counties only had two years of data available, thus were not captured in the calculation of percent change:
 - **Durham:** percent change -21.7%
 - **Nash:** percent change -15.2%



- Interestingly, despite Forsyth having the third highest overall avg readmission rate, this county actually shows a decline in percent change over the years.
- When looking at regions, Eastern North Carolina had the highest average percent change in readmission rates, followed by Western North Carolina





PERCENT OF RESIDENTS EXPERIENCING ONE OR MORE FALLS WITH MAJOR INJURY

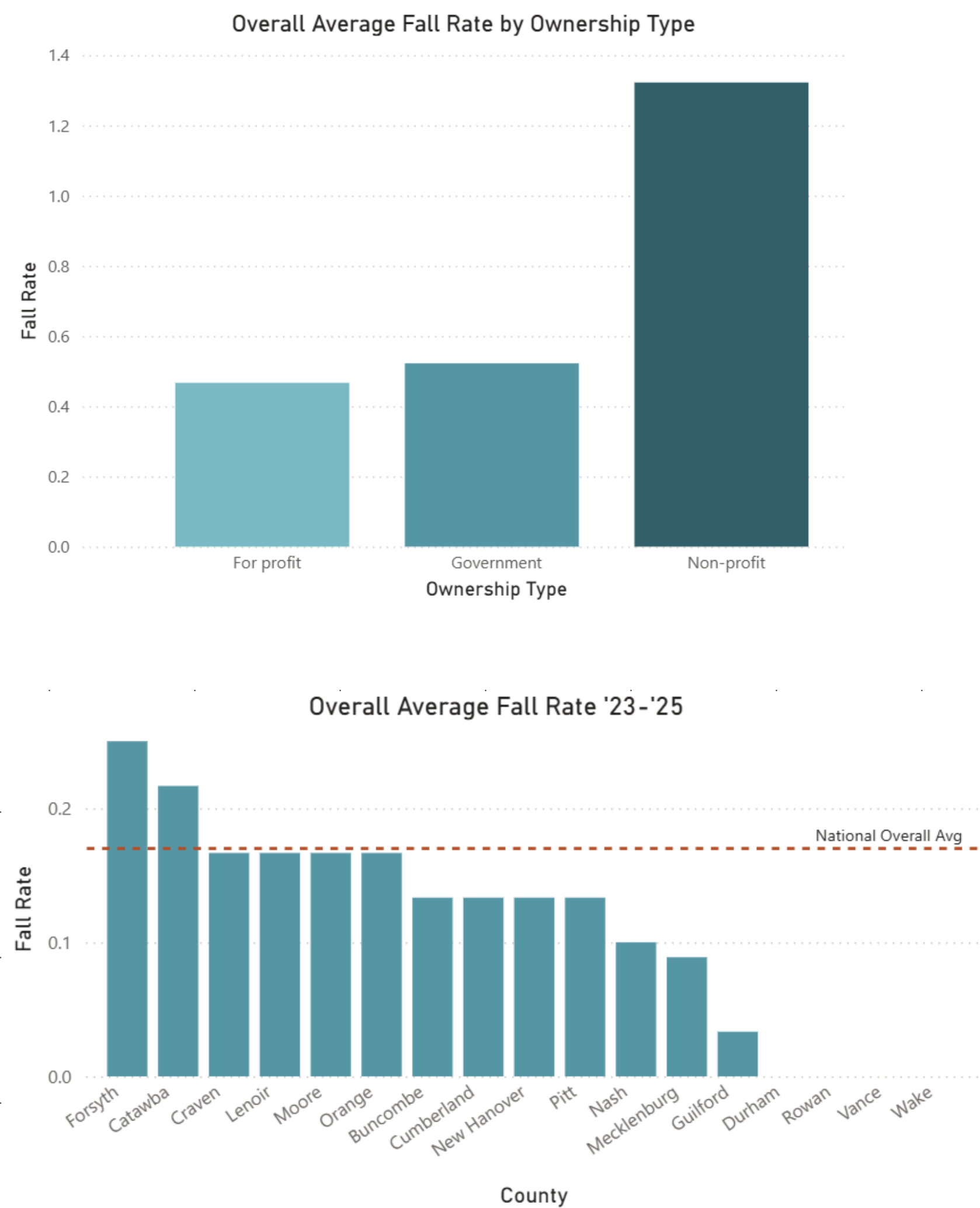
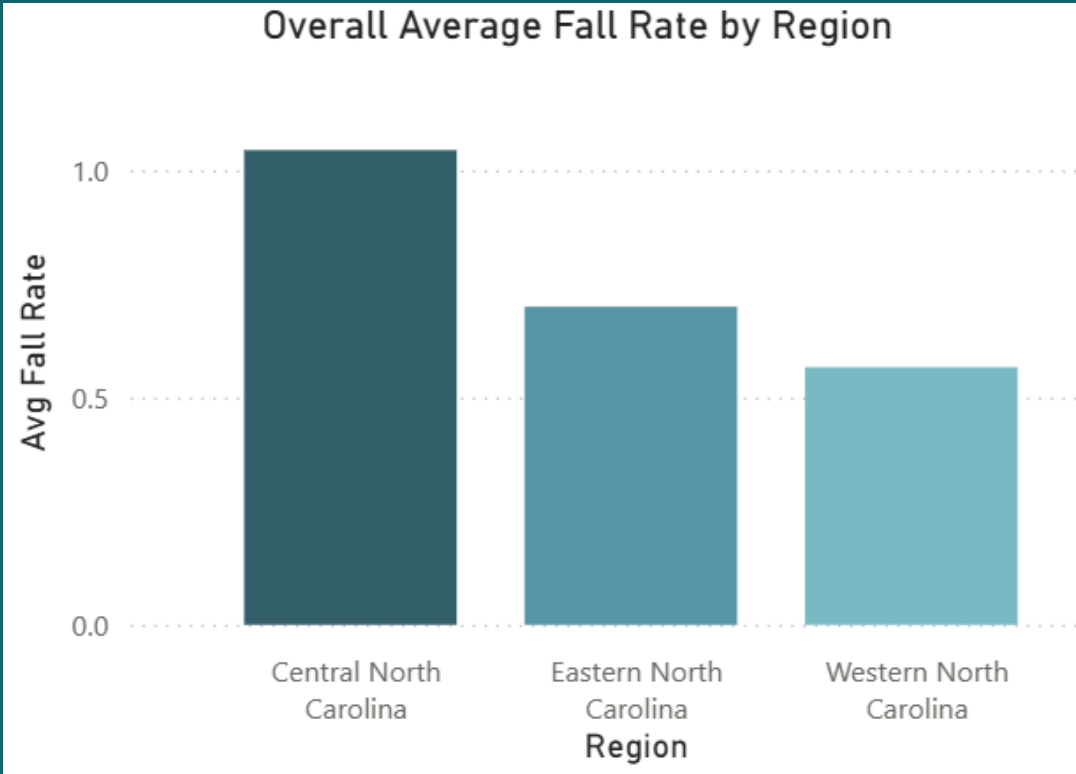
CMS ID:I022.01

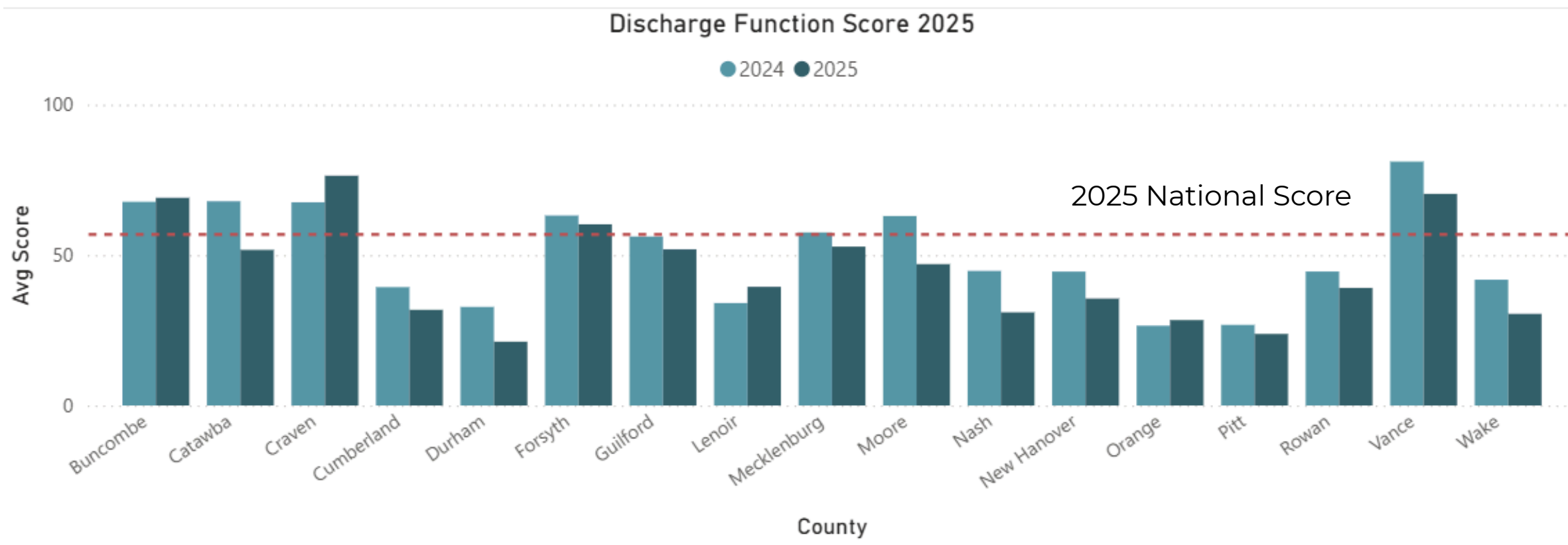
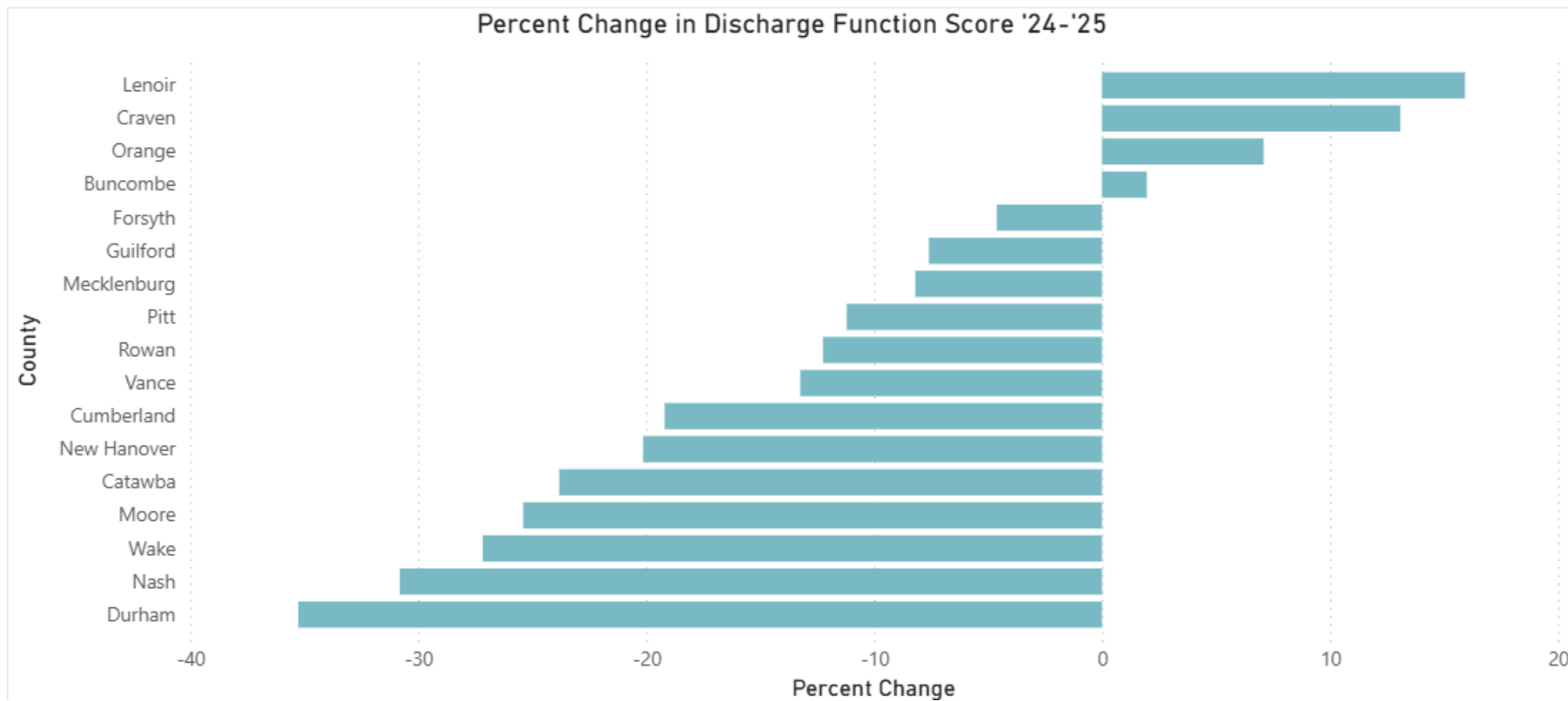
- This measure looks at the percent of IRF patients who experience one or more falls with major injury during their IRF stay
- Overall rate of falls with a major injury has increased over the last three years both nationally and in North Carolina.
- Nevertheless, North Carolina’s fall rate is consistently below the national average.
- Percent Change NC: 183.3%
- Percent Change Nationally: 100%

PERCENT OF RESIDENTS EXPERIENCING ONE OR MORE FALLS WITH MAJOR INJURY

CMS ID:I022.01

- Looking at ownership types, non-profits by far stand out with the highest overall average rate of falls.
- Forsyth and Catawba counties had average fall rates well above the national average.
- Central North Carolina also had the highest overall average fall rate of the three regions.

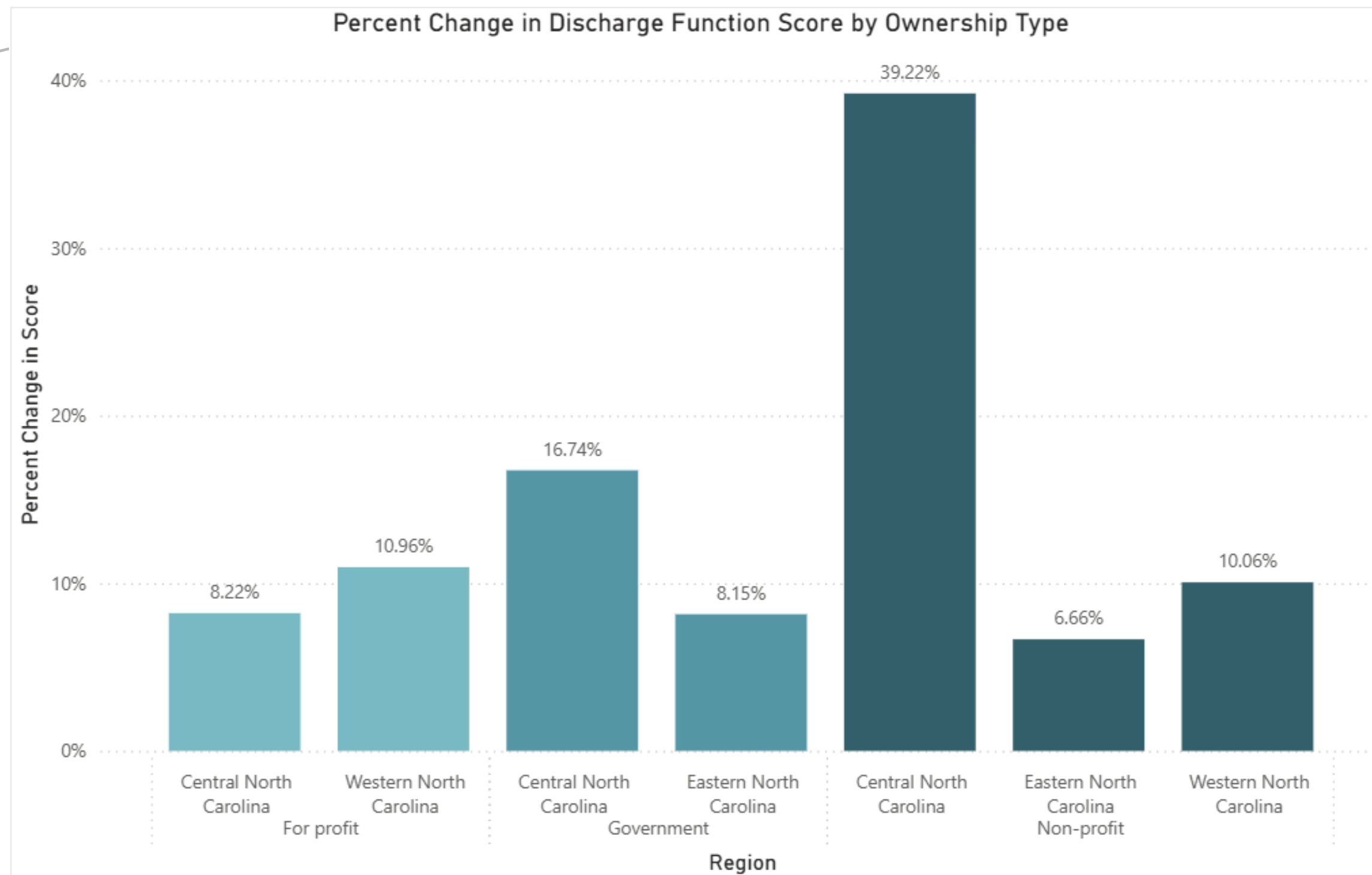




DISCHARGE FUNCTION SCORE

(CMS ID: 1026.02)

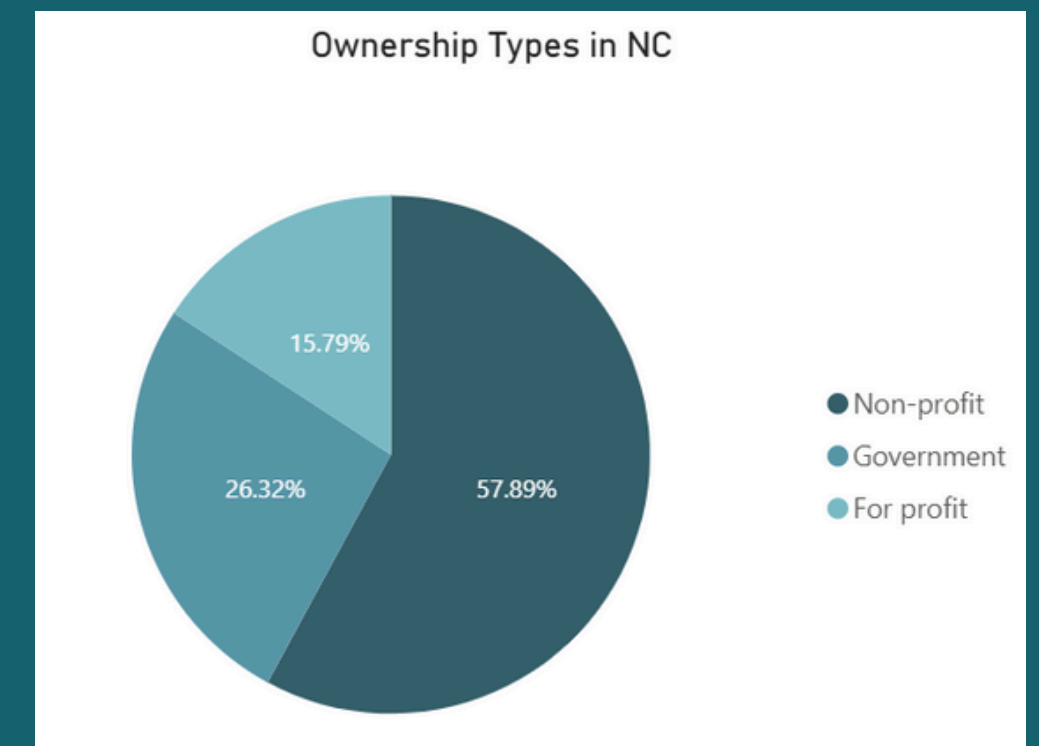
- Percent of patients who are at or above an expected ability to care for themselves and move around at d/c. Higher is better.
- This measure was introduced in 2023, thus reporting of collected data was not available until September 2024.
- National Data was only available for 2025
- 53% of NC IRFs performed lower than the national average
- 76% of counties showed a decrease in Discharge Function Scores from '24-'25



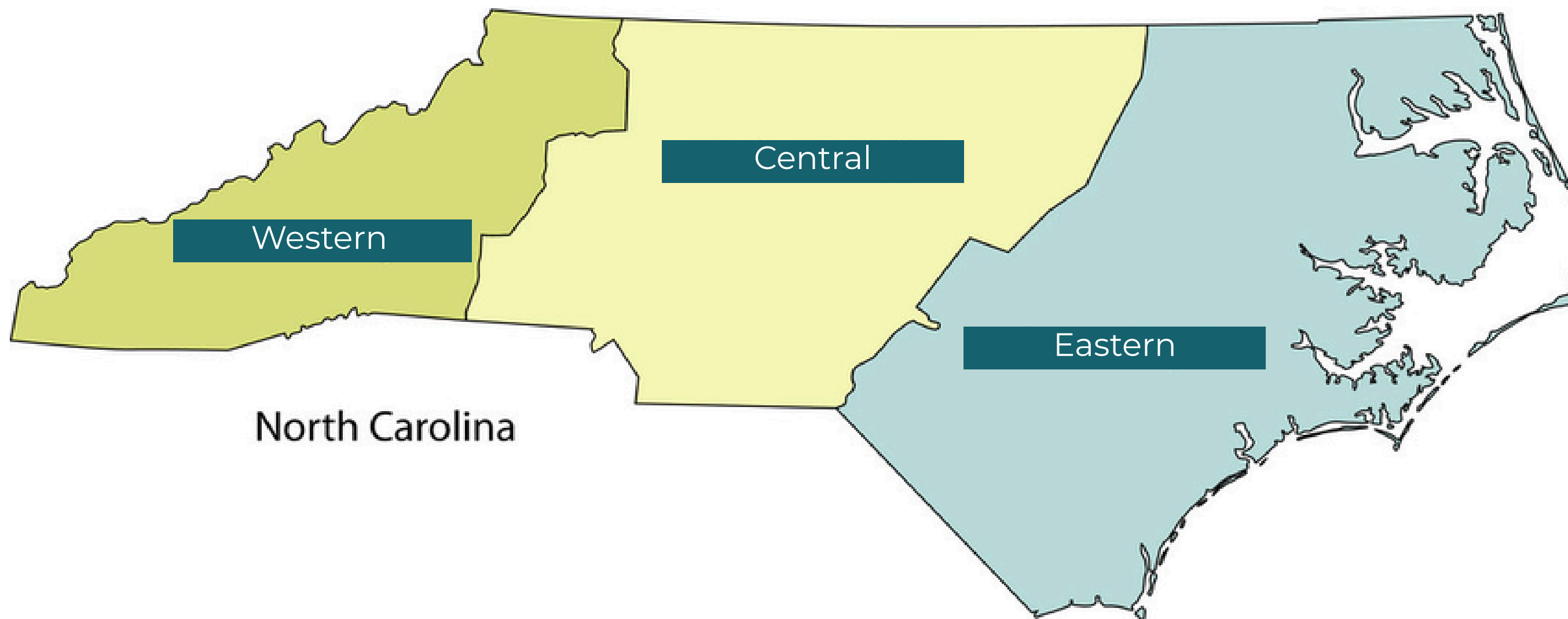
DISCHARGE FUNCTION SCORE

(CMS ID: 1026.02)

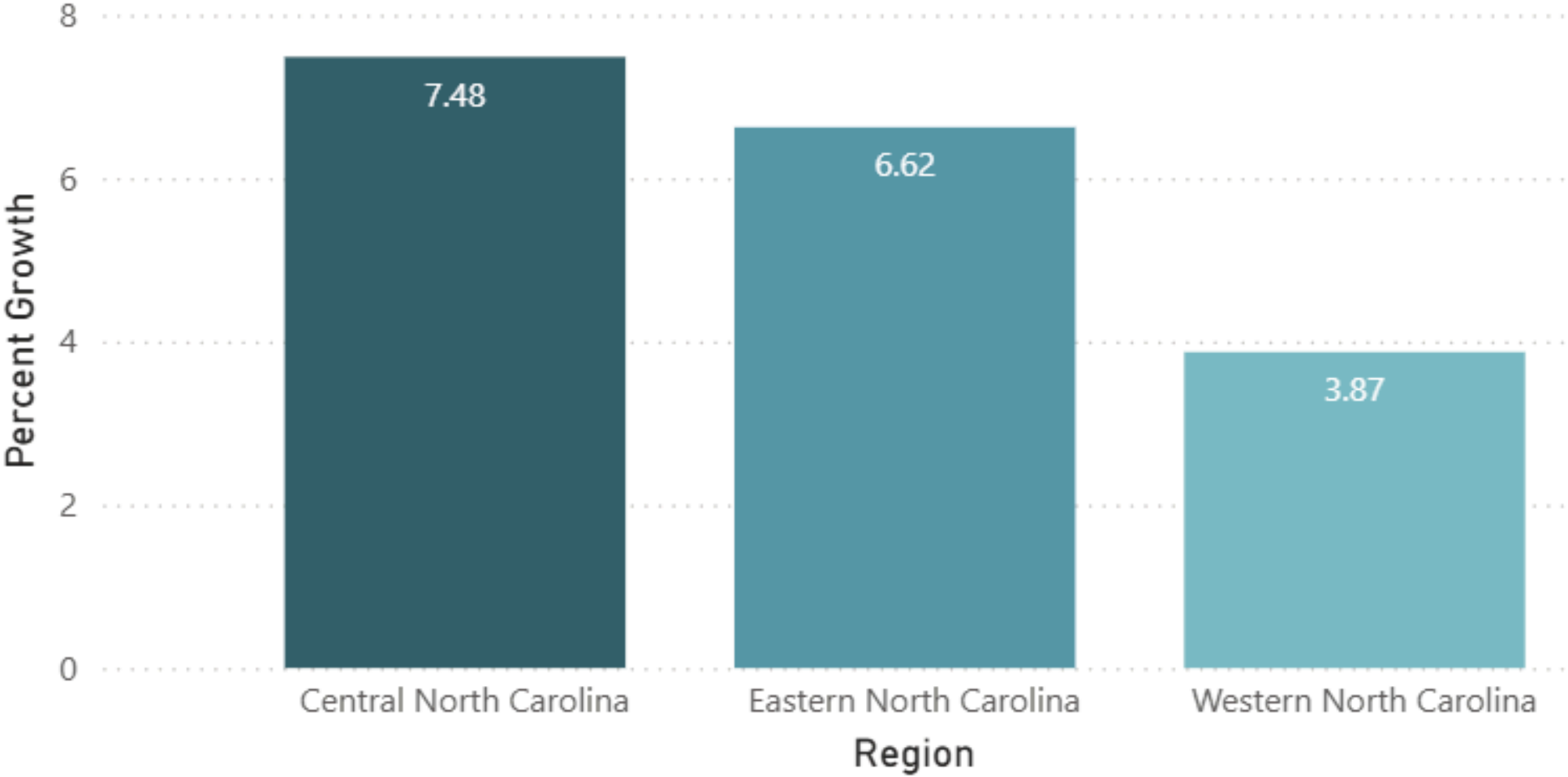
- Non-profit organizations appear to have a higher overall percent in change of the Discharge Function Score.
- Non-Profits are also the most common organization type in NC.
- Eastern North Carolina had the lowest percent change in score.



REGION



Percent Growth of 65+ 2023-2025

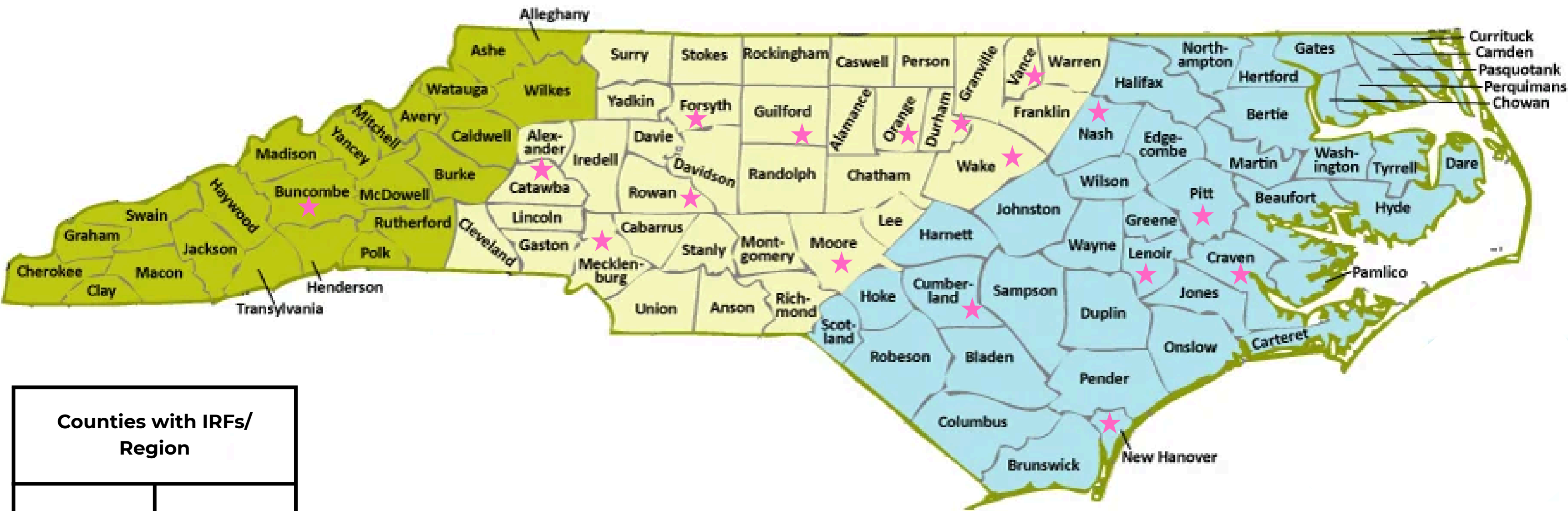


Region	2023 Percent 65 Plus	2024 Percent 65 Plus	2025 Percent 65 Plus
Central	14.6%	15%	15%
Eastern	16.2%	16.5%	16.7%
Western	20.7%	21.1%	21.5%

REGIONAL GROWTH

- As expected, the percent of the population aged 65 and older continues to grow year by year.
- Central North Carolina shows the highest percent of growth of the three regions.
- Despite being slowest in terms of growth, Western North Carolina is home to the highest percent of ages 65 and up.

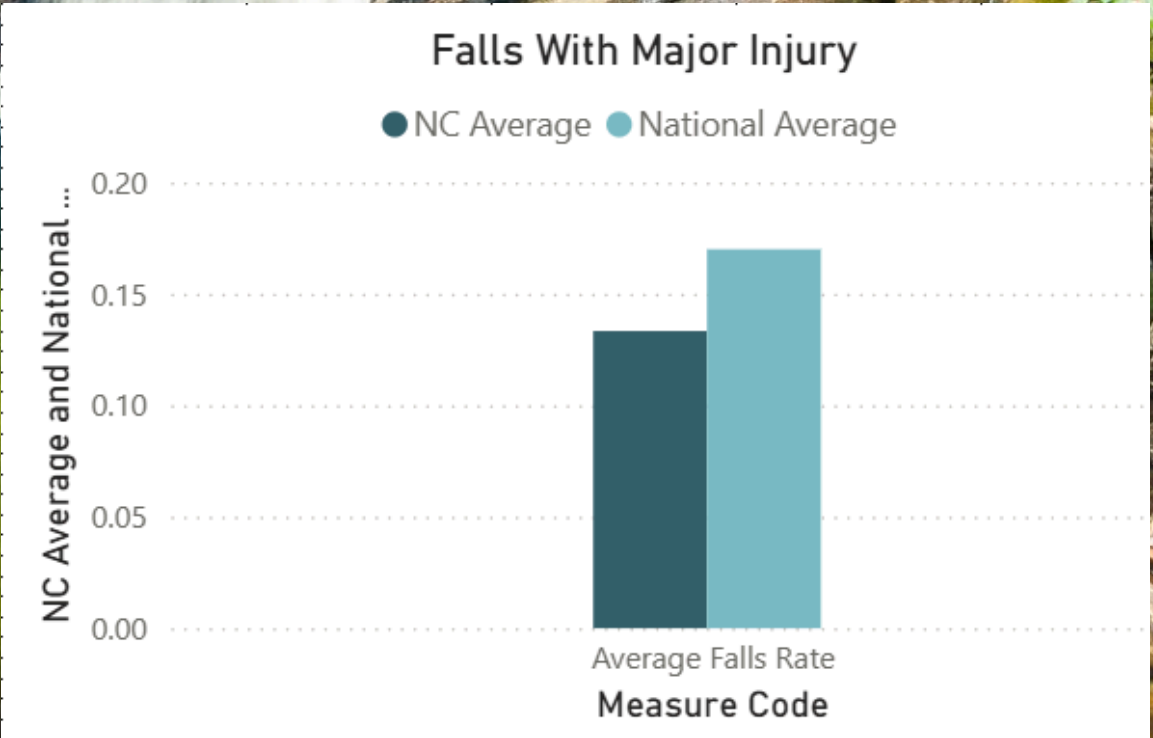
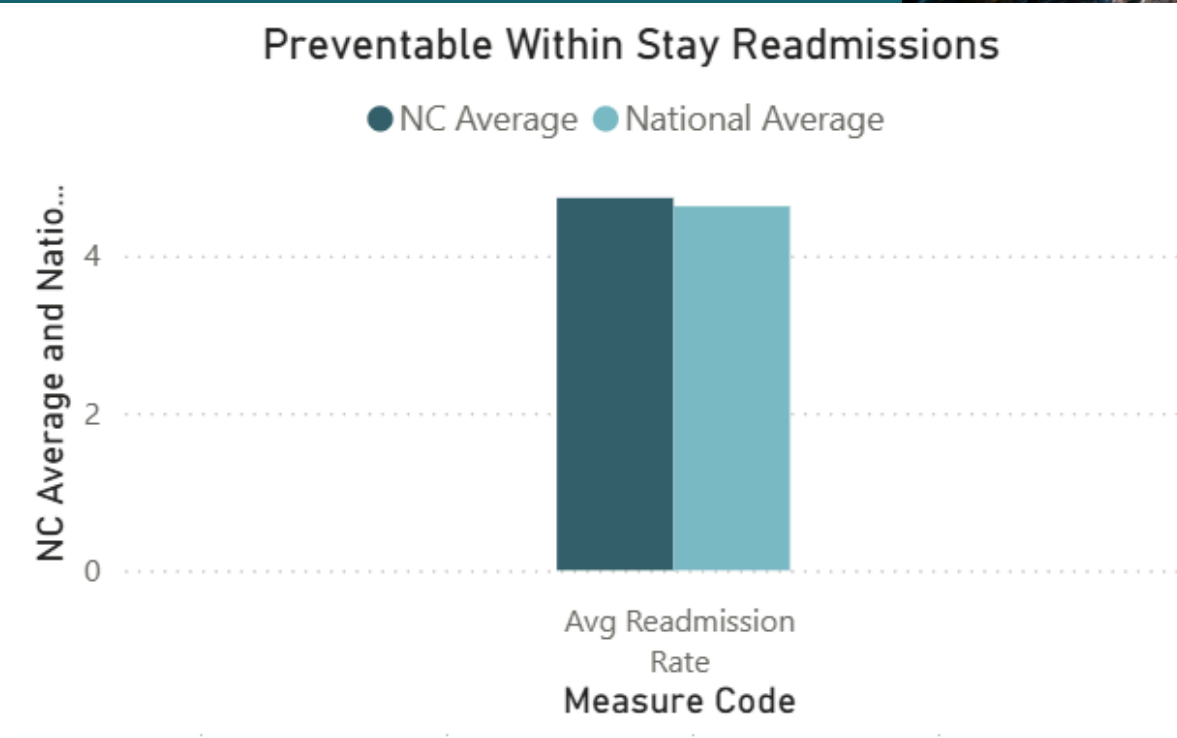
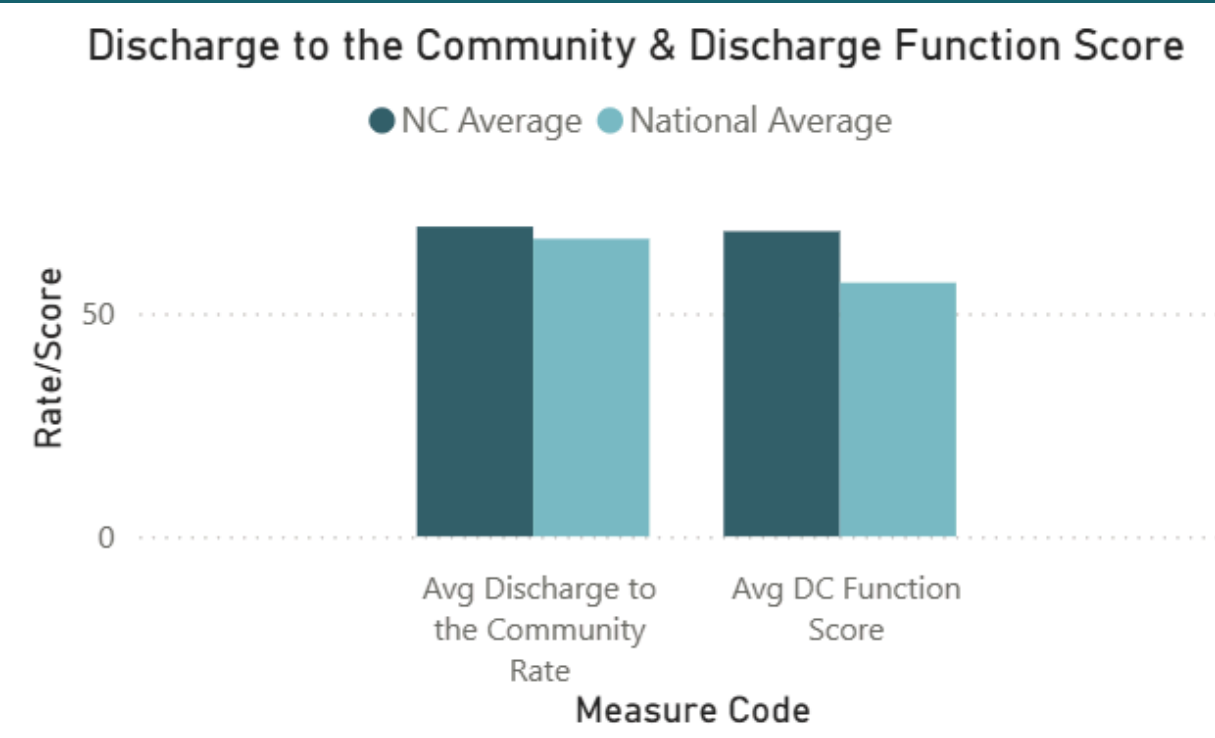
Location of Inpatient Rehab Facilities



Counties with IRFs/ Region	
Western	1
Central	10
Eastern	6

BUNCOMBE COUNTY INPATIENT REHAB HOSPITAL

- Here we examine Buncombe county's overall average score for the four quality measures over the last three years compared to the national average.
- Buncombe County outperformed the national average in measures of average discharge to the community, discharge function, and falls with major injury. Readmission rates, however, were slightly above the national average.
- This study suggests that Buncombe County's IRF provides safe, effective, and overall good quality of care for its patients.



KEY TAKEAWAYS

01. Quality

- North Carolina consistently outperforms national averages on key quality measures, with higher rates of discharge to the community and lower rates of hospital readmissions and falls resulting in major injury.

02. Ownership Type

- While data suggests that non-profit IRFs tend to have higher rates of readmissions and falls with major injury, it's important to recognize that non-profits represent the majority of IRFs in North Carolina. This larger sample size may contribute to the higher reported rates due to the broader data pool.

03. Regional

- Eastern North Carolina consistently underperforms on several key quality measures, including discharge to the community, readmission rates, and falls with major injury, warranting further investigation into potential contributing factors.
- Western North Carolina has the highest percentage of residents aged 65 and older, yet the region is served by only one IRF. This disparity suggests a potential need for additional rehabilitation resources to meet the growing demands of the aging population.



CONSIDERATIONS

- Additional time and data are needed to conduct an accurate analysis of North Carolina's performance on the Discharge Function Score, as the measure is newly introduced and lacks historical depth.
- Central North Carolina consistently performs well across many quality measures, likely due in part to its status as the most populous region in the state. A larger population base allows for a more robust and representative dataset.
- Western North Carolina may warrant further exploration to determine whether the region could support additional inpatient rehabilitation facilities, especially in light of demographic trends and access to care.
- Eastern North Carolina would benefit from a more in-depth analysis to better understand the underlying factors contributing to its consistently lower scores on reported quality measures. Identifying potential barriers or gaps in care could help inform targeted improvements.



THANK YOU!



vgcarter2014@gmail.com



[LinkedIn](#)



[GitHub](#)